

DataWeave - Declarative Data Pipeline Framework

Maharshi Soni | data-engineering | advanced

Description

A pip-installable data pipeline framework with YAML-defined ETL workflows. Built-in operators for transform, validate, profile, and load data with HTML profiling reports.

Tech Stack

pandas, pydantic, pyyaml, click, jinja2, pytest

Features

- YAML-defined pipeline configuration
- Built-in operators: filter, transform, aggregate, join, validate
- Automatic schema inference
- Data quality checks
- HTML profiling report generation

Sample Input

CSV data and YAML pipeline config

```
dataweave run --pipeline pipeline.yaml --input data.csv
```

Sample Output

Transformed data and quality report

```
dataweave profile --input data.csv --output report.html
```

Project Files

```
.gitignore
project-idea.json
pyproject.toml
README.md
.github/workflows/test.yml
samples/pipeline.yaml
samples/sales_data.csv
samples/validation_rules.yaml
src/dataweave/cli.py
src/dataweave/engine.py
src/dataweave/loader.py
src/dataweave/models.py
```

```
src/dataweave/profiler.py
src/dataweave/py.typed
src/dataweave/report.py
src/dataweave/schema.py
src/dataweave/__init__.py
src/dataweave/__version__.py
src/dataweave/operators/base.py
src/dataweave/operators/core.py
src/dataweave/operators/__init__.py
src/dataweave.egg-info/dependency_links.txt
src/dataweave.egg-info/entry_points.txt
src/dataweave.egg-info/PKG-INFO
src/dataweave.egg-info/requires.txt
src/dataweave.egg-info/SOURCES.txt
src/dataweave.egg-info/top_level.txt
tests/conftest.py
tests/test_cli.py
tests/test_engine.py
tests/test_models.py
tests/test_operators.py
tests/test_profiler.py
tests/test_schema.py
tests/__init__.py
```

Test Results

PASSED

Run: 57 | Passed: 57 | Failed: 0

Maharshi Soni | MIT License